

ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)
ORGANISATION OF ISLAMIC COOPERATION (OIC)
DEPARTMENT OF NATURAL SCIENCES

Final Examination
Course Number: PHY 4143
Course Title: Physics II

Winter Semester: 2024-2025
Full Marks: 120
Time: 2 Hours

Please answer according to the order of the questions. Answer all the 4 (FOUR) questions. The symbols have their usual meanings. Marks of each question and the corresponding CO and PO are written in the brackets.

1. a) Define resonance and power factor of a series R-L-C circuit.

(06)

(CO1)

(PO1)

- b) Illustrate the expression for the impedance of a series R-L-C circuit and draw the corresponding phasor diagram, stating the conditions under which the circuit behaves as inductive or capacitive.

(14)

(CO2)

(PO1)

- c) For the circuit shown in Fig. 1

(10)

i. Calculate the total impedance Z_T in polar form,

(CO3)

ii. Draw the impedance diagram.

(PO2)

iii. Find the current i in phasor form.

Given, $e = 6 \sin(314t + 60^\circ)$

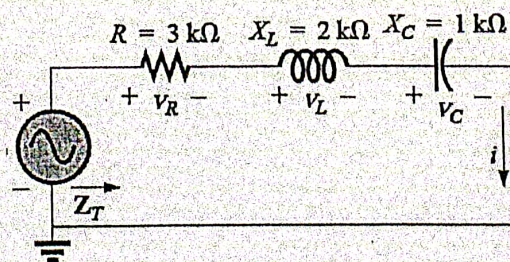


Fig. 1

2. a) Define magnetic flux density and magnetic intensity and write the relation between them.

06

(CO1)

(PO1)

- b) Describe and explain the expressions for magnetic force on a moving charge as well as a current carrying conductor.

14

(CO2)

(PO2)

- c) Apply Thevenin's theorem to find the Thevenin equivalent circuit for the network external to resistor R in Fig. 2 and determine the current flowing through R,

10

(CO3)

(PO2)

Given $E = 10 \text{ V} < 0$,

$X_L = 8 \Omega$,

$X_C = 2 \Omega$,

and $R = 10 \Omega$.

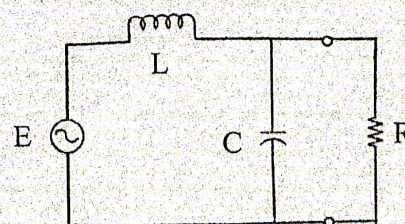


Fig. 2

3. a) State Biot-Savart's law and Ampere's law.

08

- b) Express the value and direction of magnetic field due to steady current in a long straight wire using Biot-Savart's law. *Don't ignore*
- c) A solenoid of 1.33 m long and 2.60 cm in diameter carries a current of 17.8 A. The magnetic field inside the solenoid is 22.4 mT. Find the length of the wire forming the solenoid. *B, I*
4. a) Explain the hysteresis curve in detail with a neat labeled diagram for soft and hard magnetic materials and discuss its significance. *less energy loss*
- b) Define hysteresis loss and eddy current loss. How can they be minimized?
- c) Determine the value of I required to establish magnetic flux of $\Phi = 0.75 \times 10^{-4}$ Wb in the series magnetic circuit of Fig. 3. Given the value of H is 280 At/m for cast steel.

